



Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ APT: Arbitrage Pricing Theory
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HJ: Hansen–Jagannathan (distance)
- ▣ HML: High Minus Low (value factor)
- ▣ IPCA: Instrumented Principal Component Analysis
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MOM: MOMentum factor
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares

Acronyms used in this material (2/2)

- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ RF: Risk-Free rate
- ▣ RMSE: Root Mean Squared Error
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROE: Return On Equity
- ▣ RP-PCA: Risk-Premium Principal Component Analysis
- ▣ SDF: Stochastic Discount Factor
- ▣ SMB: Small Minus Big (size factor)
- ▣ SML: Security Market Line
- ▣ UMD: Up Minus Down (momentum factor)
- ▣ US: United States

Outline

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What Is a Factor Model?

Definition

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \dots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k ; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock

Key message of this chapter

- Expected returns should be explained by **exposures to priced risks**, not by the name of the asset; testing this claim is the core of empirical asset pricing

Why Factor Models Matter

- ▣ **Cost of capital:** the return investors require from a firm or project
- ▣ **Performance evaluation:** did a fund beat the market, or did it only carry more risk?
- ▣ **Risk management:** N assets need $N(N + 1)/2$ covariances; K factors reduce this drastically
- ▣ **Portfolio construction:** factor exposures can be bought directly (factor ETFs, Section 8)
- ▣ **Asset pricing:** which risks are rewarded, and why? (textbook treatment: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 5–6)

Data Used in This Chapter

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size $\times$ B/M portfolios	US stocks sorted on two characteristics	1963–2026
US sector ETFs	11 sectors of the S&P 500	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 Romanian stocks and their index	2015–2026

- Factor and portfolio returns: [Kenneth French Data Library](#)
- Prices: repository folder data/market, up to 18 September 2026

The Data

- ▣ **ETFs** used in the chapter
 - ▶ Sector ETFs: XLK (technology), XLF (financials), XLE (energy), XLU (utilities), ...; SPY tracks the S&P 500
- ▣ **B/M**: book-to-market ratio, accounting value of equity divided by its market value
- ▣ **BET**: the benchmark index of the BVB
- ▣ Blue chips used: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL (full names on the chart)
- ▣ ETFs and stocks: dividend- and split-adjusted prices; indices: closing values (Chapter 1)

Excess Returns and the Risk-Free Rate

- **Risk-free rate** $R_{f,t}$: return on an asset with no default risk over the period
 - ▶ US: one-month Treasury bill rate (column RF of the Fama–French files); mean 4.35% a year since 1963
- **Excess return**: $R_{i,t}^e = R_{i,t} - R_{f,t}$, the reward for bearing risk
- **Market excess return** (Mkt-RF): value-weighted US market minus $R_{f,t}$
- Assumption for the BVB: daily $R_{f,t}$ is treated as constant
 - ▶ A constant cancels in covariances, so betas are unchanged; only the intercept shifts
- Regressions on Fama–French factors use simple returns (the factors are simple returns); rolling, BVB and PCA analyses use log returns (Chapter 1)

Prerequisites Recap: Mean–Variance and the Tangency Portfolio

- **Markowitz (1952)**: with weights w , expected excess returns μ and covariance Σ , the frontier solves

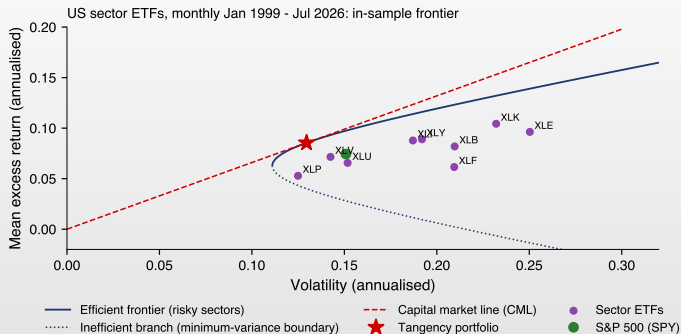
$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, \quad w^\top \mathbf{1} = 1$$

- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f] / \sigma_p$; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$w_{\text{tan}} = \Sigma^{-1} \mu / \mathbf{1}^\top \Sigma^{-1} \mu$$

- **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\text{tan}} \sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar A1; estimation error and shrinkage: Chapter 4

The Efficient Frontier of US Sectors



□ Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)

Reading the Frontier

- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49
 - ▶ Here the market is **not** on the frontier, as the CAPM would require
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- Caveat: the frontier is **in-sample**; estimated means are very noisy
 - ▶ Out-of-sample optimisation and covariance shrinkage: Chapter 4

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics
1990

Photo: Larry D. Moore (2007); CC BY 4.0;
[Wikimedia Commons](#)

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- Question: if every investor holds a mean–variance efficient portfolio, what are equilibrium prices?
- Answer: everyone holds the same tangency portfolio, so it must be the **market portfolio**
- [Black \(1972\)](#): a version without a risk-free asset (the zero-beta CAPM)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

The CAPM Equation

- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i : sensitivity of asset i to the market
- Main assumptions
 - ▶ Mean–variance investors with identical beliefs, one period
 - ▶ Unlimited borrowing and lending at R_f ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns
- SML** (Security Market Line): every asset lies on the line (beta, mean) with slope equal to the premium; the CML contains only efficient portfolios. The SML is the testable statement (Section 4)

Recap: From Markowitz to the CAPM

- Mean–variance investors hold the tangency portfolio combined with the risk-free asset
- In equilibrium this portfolio is the market, and expected excess returns are β_i times the premium
- CML: efficient portfolios in (volatility, mean); SML: every asset in (beta, mean)
- Only systematic risk is rewarded under the CAPM
- In our sector data the market proxy is far from the in-sample frontier
- CAPM cost of capital needs an estimated beta and an estimated premium: the error of the premium dominates

Prerequisites Recap: the Market Model and HAC Errors

- ▣ Market model $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\text{Cov}}(R_i^e, R_m^e) / \widehat{\text{Var}}(R_m^e)$
- ▣ Variance split $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
- ▣ Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** errors, [Newey & West \(1987\)](#), with Bartlett weights
- ▣ Lag rule $L = \lfloor 4(T/100)^{2/9} \rfloor$: $T = 331$ months gives $L = 5$; $T = 757$ gives $L = 6$
- ▣ Same point estimates as OLS, only the standard errors change

Worked Example: Beta of the Technology ETF

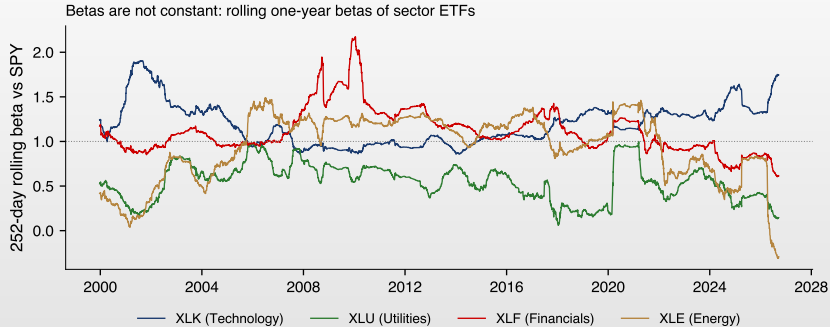
	Estimate	Classical s.e.	Newey–West s.e.
α (annualised)	0.33%	2.27%	2.45%
β	1.277	0.041	0.082

- ▣ XLK on the US market factor Mkt-RF, monthly excess returns 1999–2026, $T = 331$, $R^2 = 0.742$
- ▣ Newey–West with $L = \lfloor 4(3.31)^{2/9} \rfloor = \lfloor 5.2 \rfloor = 5$ lags
- ▣ The robust standard error of beta is **twice** the classical one
- ▣ Alpha is not statistically distinguishable from zero: $t = 0.33/2.45 = 0.13$

Estimation Choices in Practice

- ▣ **Frequency:** daily data give more observations but suffer from asynchronous trading
- ▣ **Window:** 3–5 years of monthly data or 1–2 years of daily data are common
- ▣ **Market proxy:** S&P 500 (SPY) for US assets, BET for Romanian stocks
- ▣ **Data cleaning:** unadjusted capital events create fake returns
 - ▶ Example: a daily log return of -6.6 for DIGI in May 2017; such days are removed
- ▣ Always report the sample, the frequency and the standard error of beta

Betas Are Not Constant



- Rolling beta: OLS beta on the last 252 daily log returns, recomputed every day

Reading the Rolling Betas

- ▣ Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- ▣ Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end of the sample
- ▣ Energy (XLE): from -0.31 to 1.49; negative at the end, when oil moved against the market
- ▣ Utilities (XLU): from 0.06 to 1.00; the classic defensive sector
- ▣ A single full-sample beta hides these regimes; conditional models: Chapters 5–6

Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is

$$\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$$

- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

- ▣ β_i^V is the posterior mean under a Normal prior $\beta_i \sim N(\bar{\beta}, \sigma_{\beta}^2)$ and $\hat{\beta}_i | \beta_i \sim N(\beta_i, \text{se}_i^2)$ (Seminar A3)
- ▣ σ_{β}^2 is the variance of the **true** betas: $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$

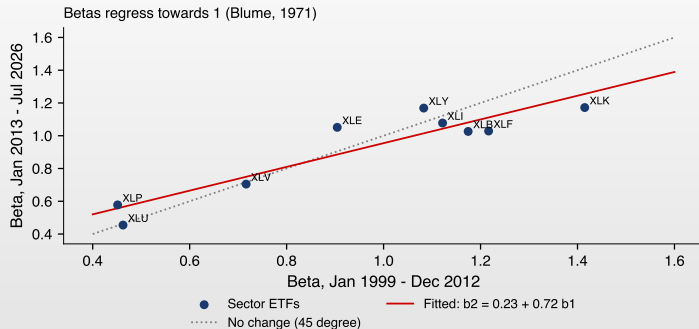
Worked Example: Vasicek Beta of XLK

- ▣ Sample January 1999 – December 2012, monthly, 9 sector ETFs
- ▣ XLK: $\hat{\beta} = 1.415$, $\text{se}(\hat{\beta}) = 0.139$ (Newey–West)
- ▣ Prior mean $\bar{\beta} = 0.949$; prior variance $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$
 - ▶ 0.116: variance of the 9 estimated betas; 0.008: their average squared standard error
- ▣ Weight: $w = 0.108 / (0.108 + 0.139^2) = 0.849$

$$\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$$

- ▣ The realised beta in 2013–2026 was 1.172: shrinkage moved the estimate in the right direction

Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)

Reading the Beta Regression

- ▣ Fitted line: $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67 \hat{\beta}$
- ▣ **RMSE** (Root Mean Squared Error) of the forecast of later betas

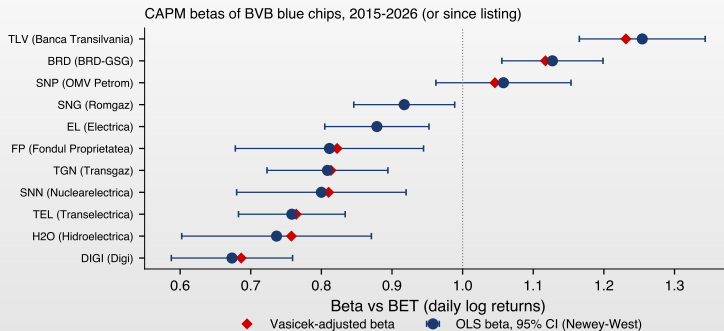
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$$

- ▣ Raw beta: 0.135; Blume rule: 0.109; Vasicek: 0.114
- ▣ Shrinkage cuts the forecast error by 15–20%
- ▣ Only 9 sectors: the lesson is qualitative

Romania: Betas of the BVB Blue Chips

- ▣ Regression of daily log returns of each stock on daily log returns of BET
- ▣ Sample from 6 January 2015 or from the listing date (DIGI 2017, H2O 2023)
- ▣ Risk-free rate: assumed constant at daily frequency, so it does not affect beta
- ▣ Days with $|r_t| > 0.5$ (capital events not adjusted in prices) are removed: one day for DIGI, two for FP
- ▣ Newey–West 95% confidence intervals and Vasicek-adjusted betas (empirical-Bayes prior from the 11 stocks)

CAPM Betas on the Bucharest Stock Exchange



□ Blue dots: OLS betas with 95% intervals; red diamonds: Vasicek-adjusted betas

Reading the BVB Betas

- ▣ Banks are the most market-sensitive: Banca Transilvania 1.25, BRD 1.13
- ▣ OMV Petrom 1.06; Romgaz 0.92; utilities and telecom below: Transelectrica 0.76, Digi 0.67
- ▣ R^2 : 0.59 for TLV and 0.52 for BRD, but only 0.21 for DIGI
- ▣ Large index weights (TLV, SNP, BRD) mechanically raise their R^2 against BET
- ▣ Vasicek changes little here: the daily estimates are precise (se $\approx$ 0.04–0.07)

Recap: Estimating Beta

- ▣ Beta is an OLS slope; report it with Newey–West standard errors
- ▣ Betas move over time: sector betas ranged from about -0.3 to 2.2 in rolling windows
- ▣ Extreme betas regress towards one: Blume and Vasicek shrinkage improve forecasts
- ▣ Romanian banks have the highest betas against BET; utilities and telecom the lowest
- ▣ Clean the data first: a single unadjusted corporate event destroys a regression

Jensen's Alpha and the GRS Test

- ▣ **Jensen's alpha** (Jensen (1968)): intercept of the market model, $\alpha_i = \mathbb{E}[R_i^e] - \beta_i \mathbb{E}[R_m^e]$; the CAPM implies $\alpha_i = 0$ for every asset
- ▣ One t -test per asset inflates false rejections (Section 7): **GRS**, Gibbons, Ross & Shanken (1989), tests $H_0 : \alpha_1 = \dots = \alpha_N = 0$ jointly

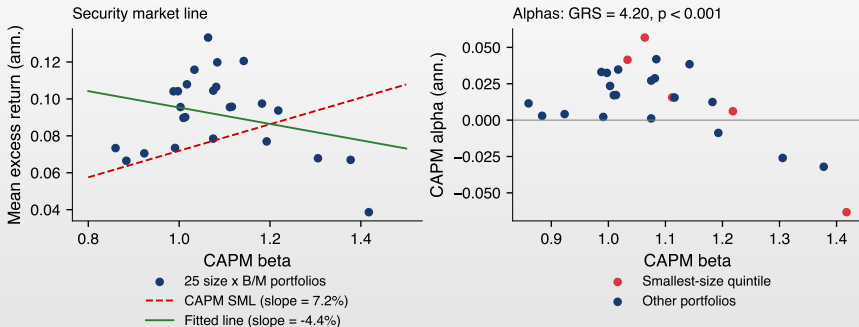
$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣ $\hat{\Sigma}$: maximum-likelihood residual covariance (divided by T); $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market
- ▣ Exact, conditional on the factors, under i.i.d. multivariate Normal errors independent of the factors, with zero mean and constant nonsingular covariance; F : Fisher distribution
- ▣ Large W : the market is far from the mean–variance frontier of the test assets

The 25 Size and Book-to-Market Portfolios

- ▣ US stocks sorted each June into 5 **size** quintiles (market value) and 5 **B/M** quintiles
- ▣ $5 \times 5 = 25$ value-weighted portfolios: from small-growth (small, low B/M) to big-value
- ▣ Standard test assets since [Fama & French \(1993\)](#): their average returns vary widely
- ▣ Monthly, July 1963 – July 2026, $T = 757$
- ▣ Portfolios instead of single stocks: betas are estimated much more precisely

The Security Market Line in the Data



□ Left: average excess returns vs CAPM betas; right: alphas, smallest-size quintile in red

Reading the Security Market Line

- ▣ The CAPM predicts a line with slope equal to the premium, 7.2% a year
- ▣ The fitted cross-sectional line has slope -4.4% : higher beta, **lower** average return
- ▣ Betas span only 0.86–1.42, while average excess returns span 3.9%–13.3%
- ▣ Small-value alpha $+5.7\%$, small-growth alpha -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- ▣ $GRS = 4.20$, p -value 8×10^{-11} : the CAPM is rejected

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\Sigma}$ and the moments estimated by maximum likelihood

$$\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2(f, R) - \widehat{SR}^2(f)$$

- $\widehat{SR}^2(f, R)$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets

- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T - N - 1}{N} \cdot \frac{\widehat{SR}^2(f, R) - \widehat{SR}^2(f)}{1 + \widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1 + 0.134^2) = 4.20$; under H_0 , $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$, so the expected gain is only $N(1 + \widehat{SR}^2(f))/(T - N - 3) = 0.035$, against 0.146 observed

When the GRS Assumptions Fail

- ▣ **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
 - ▶ GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_N^2$, with $\hat{V}_\alpha$ the HAC estimate of $\text{Var}(\hat{\alpha})$: robust to heteroskedasticity and autocorrelation
 - ▶ Residual bootstrap under H_0 : resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar B11); robust to non-Normality only if errors are i.i.d. and independent of the factors
- ▣ **Large N** : $\hat{\Sigma}$ has rank at most $T - 2$, so GRS is not computable when $N \geq T - 1$
 - ▶ Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$); power falls well before that point
- ▣ Tests for many assets that avoid $\hat{\Sigma}^{-1}$: [Pesaran & Yamagata \(2024\)](#); power enhancement by screening large alphas: [Fan, Liao & Yao \(2015\)](#)

A Flat SML: Black's Insight



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage
- High-beta assets become expensive, so their future returns are lower
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above R_f
- The same Black co-authored the Black–Scholes option formula (Chapter 12)

Betting Against Beta

- Frazzini & Pedersen (2014) turn the flat SML into a trading strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L : low-beta portfolio, leveraged to beta 1; H : high-beta portfolio, de-leveraged to beta 1
- The portfolio has beta zero by construction, yet earns a positive average return in their data
- Explanation: leverage constraints of investors, as in Black (1972)

Roll's Critique

- ▣ Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth
- ▣ Stock indices omit bonds, real estate, private firms and human capital
- ▣ Mathematical fact: an efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier
- ▣ Hence a CAPM test is a joint test of the model and of the efficiency of the proxy
- ▣ Practical answer: evaluate models by how well they price assets, not as literal truths

Low Power: GRS on Sector ETFs

- ▣ Same test on the 9 sector ETFs, monthly 1999–2026 ($T = 331$)
- ▣ $GRS = 0.53$, $p = 0.85$: the CAPM is **not** rejected; the sectors raise the annual Sharpe ratio only from 0.51 to 0.66
- ▣ Alphas: utilities +3.4%, energy +2.4%, financials −2.3% a year, all statistically weak
- ▣ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- ▣ Lesson: failing to reject is weak evidence; choose test assets with dispersion in expected returns

Recap: Testing the CAPM

- ▣ The CAPM says all alphas are zero; the GRS test checks this jointly
- ▣ On the 25 size $\times$ B/M portfolios, 1963–2026, the CAPM is strongly rejected
- ▣ The empirical SML is flat or negative; leverage constraints (Black; BAB) offer one explanation
- ▣ Roll: every test also tests the market proxy
- ▣ Test assets matter: sector ETFs cannot reject the CAPM

Cross-Sectional Tests: Fama and MacBeth



Eugene Fama receiving the Morgan Stanley–AFA
Award, 2008

Photo: [TonyTheTiger \(2008\)](#); CC BY-SA 3.0;
[Wikimedia Commons](#)

- **Fama & MacBeth (1973)**: estimate the **price of risk** λ of each factor
- **Risk premium** λ_k : extra expected return per unit of exposure $\beta_{i,k}$
- Model: $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$
- Allows factors that are **not** traded portfolios, and firm characteristics as regressors
- Fama: Nobel Prize in Economics 2013

The Two Passes

- ▣ **Pass 1** (time series): for each asset i , estimate the loadings

$$R_{i,t}^e = a_i + \beta_i^\top \mathbf{f}_t + \varepsilon_{i,t} \quad \Rightarrow \quad \hat{\beta}_i$$

- ▣ **Pass 2** (cross section): for each month t , regress returns on the estimated betas

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \quad i = 1, \dots, N$$

- ▣ Estimate: $\hat{\boldsymbol{\lambda}} = \frac{1}{T} \sum_t \hat{\boldsymbol{\lambda}}_t$, the average of T monthly slopes
- ▣ λ_0 : pricing error common to all assets; zero if the model is right

Fama–MacBeth Standard Errors

- Treat the monthly estimates $\hat{\lambda}_{k,t}$ as a sample of size T

$$se_{FM}(\hat{\lambda}_k) = \frac{s(\hat{\lambda}_{k,t})}{\sqrt{T}}, \quad s^2 = \frac{1}{T-1} \sum_t (\hat{\lambda}_{k,t} - \hat{\lambda}_k)^2$$

- Advantage: automatically robust to cross-sectional correlation of residuals
- If $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series instead
- Problem: the betas of Pass 2 are **estimated**, not known

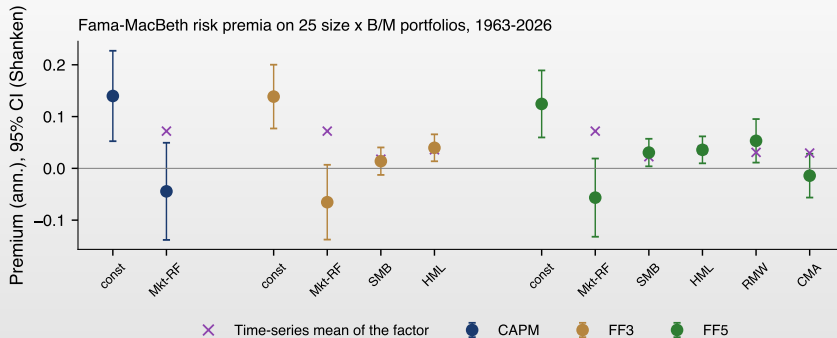
Errors in Variables: the Shanken Correction

- ▣ **Errors in variables:** using $\hat{\beta}$ instead of β makes se_{FM} too small
- ▣ The Fama–MacBeth variance already contains the factor term Σ_f/T ; [Shanken \(1992\)](#) scales only the rest by $(1 + c)$

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \left[\text{Var}_{FM}(\hat{\lambda}) - \frac{1}{T} \Sigma_f \right] + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

- ▣ Σ_f : covariance matrix of the factors; c : squared Sharpe ratio of the premia per period
- ▣ Small for FF3 here ($c = 0.03$ per month); potentially large when the estimated premia are large relative to the factor covariance, as for some non-traded factors; rescaling a factor leaves c unchanged
- ▣ Same result from **GMM** (Generalised Method of Moments, [Hansen \(1982\)](#)); see Quantinar

Estimated Risk Premia



□ 25 size × B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means

Reading the Risk Premia

- ▣ CAPM: intercept 14.0% (se 4.4%), market premium -4.4% (se 4.8%)
- ▣ FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); the market premium stays negative
- ▣ FF5: RMW 5.3% (2.1%); CMA -1.4% (2.2%) is not statistically distinguishable from zero (95% interval about $[-5.7\%, 2.9\%]$)
- ▣ For a traded factor, λ should equal the time-series factor mean (the crosses)
 - ▶ HML: 4.0% vs mean 3.6%, close; market: -6.5% vs mean 7.2%, far
- ▣ Market betas span only 0.86–1.42 (cross-sectional s.d. 0.14): λ_m is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term ([Kan, Robotti & Shanken \(2013\)](#))
- Write both passes as one exactly identified GMM system; the sandwich variance with the full Jacobian is the robust one
 - ▶ Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of [Jagannathan & Wang \(1998\)](#)
- FF5 on the 25 portfolios: s.e. of RMW 2.1% $\rightarrow$ 2.6%, CMA 2.2% $\rightarrow$ 2.7%, intercept 3.3% $\rightarrow$ 3.8%
- RMW: t falls from 2.5 to 2.1; HML changes by less than 0.05 percentage points

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	s.e.	t	Significant at 5%?
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- Shanken barely moves the error: $c = 0.03$ per month
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- At the $t > 3$ hurdle of Section 7, HML is borderline under every choice (2.45–3.00)

Useless and Weak Factors

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$ is pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t -test over-rejects when the model is misspecified ([Kan & Zhang \(1999\)](#))
- Simulation on our 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
 - ▶ $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
 - ▶ First-pass Wald test of $\beta_{i,g} = 0$ for all i : rejected in 6.7%; run it before pass 2
- **Weak factors** (little dispersion in betas, like the market here): identification-robust tests ([Kleibergen \(2009\)](#)); with many assets, consistency of pass 2 depends on factor strength ([Anatolyev & Mikusheva \(2022\)](#))

A Skeptical Appraisal: Cross-Sectional R^2

- The 25 size $\times$ B/M portfolios have a strong three-factor structure
 - ▶ Any factors correlated with SMB and HML fit their average returns: a high OLS R^2 is a low hurdle (Lewellen, Nagel & Shanken (2010))
- Prescriptions of Lewellen, Nagel & Shanken (2010)
 - ▶ Add test assets with other characteristics, for example 30 industry portfolios
 - ▶ For traded factors impose $\lambda = \mathbb{E}[f]$ (the factor prices itself)
 - ▶ Report the **GLS** (Generalised Least Squares) R^2 , weighted by $\hat{V}^{-1}$ like GRS, and confidence intervals for R^2
- Individual stocks instead of portfolios: more dispersion in betas, but larger errors in variables (Ang, Liu & Schwarz (2020))

Worked Example: Adding 30 Industry Portfolios

Model	OLS R^2		GLS R^2	
	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

- Monthly 1963–2026, free intercept and premia; $\hat{V}$: sample covariance of the test-asset returns
- FF3 falls from 66% to 22% once industries are added; the GLS R^2 never exceeds 26%
- With $\lambda = \mathbb{E}[f]$ and no intercept, the FF3 R^2 on the 55 assets is negative (-0.27)

Recap: Fama–MacBeth

- ▣ Pass 1 estimates betas; Pass 2 averages monthly cross-sectional slopes
- ▣ Fama–MacBeth errors handle cross-sectional correlation; Newey–West handles autocorrelation
- ▣ Shanken corrects for estimated betas (small for monthly traded factors); Kan–Robotti–Shanken also for misspecification
- ▣ Useless factors look priced in pass 2: test the first-pass betas first
- ▣ On the 25 portfolios the value and profitability premia are significant, but the cross-sectional R^2 collapses once industries are added

Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a K -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣ The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread, $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French 2×3 sort: size (small/big) × characteristic (30% / 40% / 30%)
 - ▶ **SMB** (Small Minus Big): small stocks minus big stocks
 - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- Portfolios rebalanced each June using accounting data of the previous year

The Fama–French Models

- Fama & French (1992): size and B/M explain the cross section; beta does not
- **FF3** (three-factor model), Fama & French (1993)

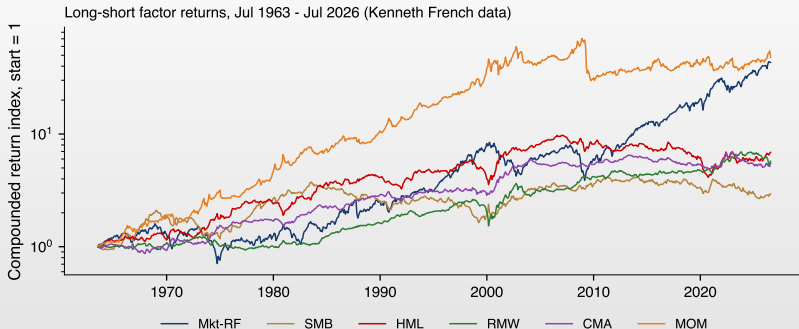
$$R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$$

- **FF5** (five-factor model), Fama & French (2015), adds
 - ▶ **RMW** (Robust Minus Weak): high minus low operating profitability
 - ▶ **CMA** (Conservative Minus Aggressive): low minus high asset growth (investment)
- FF3 uses its published SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from B/M, profitability and investment sorts, so FF3 is not nested in FF5
- Test of a model: time-series alphas close to zero, GRS not rejected

Momentum, Carhart and the q-Factor Model

- ▣ **Momentum**: past winners keep winning over 3–12 months, [Jegadeesh & Titman \(1993\)](#)
- ▣ **MOM** (UMD, Up Minus Down): stocks with high minus low return from month $t - 12$ to $t - 2$
- ▣ [Carhart \(1997\)](#) four-factor model: FF3 + MOM, standard for fund evaluation
- ▣ **q-factor model**, [Hou, Xue & Zhang \(2015\)](#): market, size, investment and profitability (ROE)
 - ▶ ROE: Return On Equity; motivated by firms' investment decisions
- ▣ FF5 and the q-model are close relatives; FF5 omits momentum

Six Factors over Sixty-Three Years

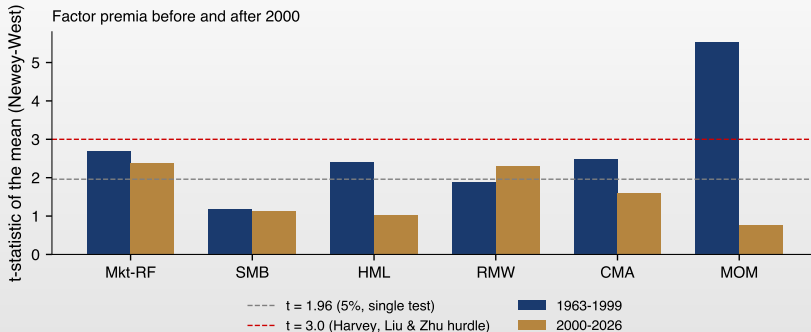


- Compounded monthly factor returns, starting at 1: excess or long-short returns, so no interest on the capital or collateral; log scale

Reading the Factor Growth

- ▣ July 1963 – July 2026: the compounded index reached 43.2 for Mkt-RF and 48.0 for MOM
- ▣ Value (HML) 6.9, profitability (RMW) 5.7, investment (CMA) 5.5, size (SMB) 2.9
- ▣ Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- ▣ Value was flat or losing from about 2007 to 2020
- ▣ Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

Do the Premia Survive?



□ t-statistic of the mean monthly factor return (Newey-West), 1963–1999 vs 2000–2026

Reading the Split-Sample t -Statistics

- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31
- Market premium robust in both halves ($t = 2.70$ and 2.39)
- McLean & Pontiff (2016): anomaly returns are 58% lower after academic publication, consistent with this decay

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in
Economics 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0;
[Wikimedia Commons](#)

- **SDF** (Stochastic Discount Factor) m_t : prices all excess returns, $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#)
- Survey of the discount-rate view: [Cochrane \(2011\)](#)
- Hansen shared the 2013 Nobel Prize with Fama and Shiller

Comparing Models: Maximum Sharpe Ratios

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(f) = \mu_f^\top \Sigma_f^{-1} \mu_f$ of its factors; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha of the added factor on the others (spanning regression)
 - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994))

Beyond Sharpe Ratios: HJ Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_b e(b)^\top \mathbb{E}[R^e R^{e\top}]^{-1} e(b), \quad e(b) = \mathbb{E}[m_t(b) R_t^e]$$

- ▶ Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$ of the previous slide; for gross returns R_t : $e = \mathbb{E}[mR] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
- ▶ Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over two quadrillion factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

Recap: Multifactor Models

- The APT justifies several priced factors but does not name them
- Factors are long-short portfolios sorted on characteristics: SMB, HML, RMW, CMA, MOM
- FF3, Carhart, FF5 and the q-model are the standard benchmarks
- Momentum and value were much weaker after 2000; profitability stayed strong
- Factor models are SDFs; compare them by the Sharpe ratio of their factors (HML adds no significant alpha to the other FF5 factors and MOM) or by the HJ distance

The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs $|t| > 3.0$

False Discovery Rate: Benjamini–Hochberg

- ▣ **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- ▣ **BH** procedure (Benjamini & Hochberg (1995)): find the largest j with

$$p_{(j)} \leq \frac{j}{M} \alpha$$

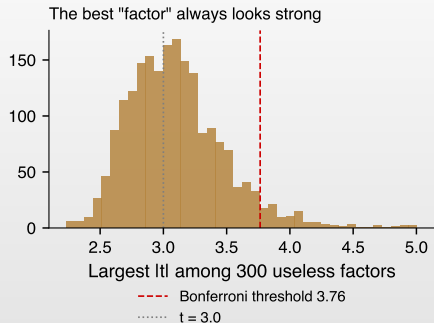
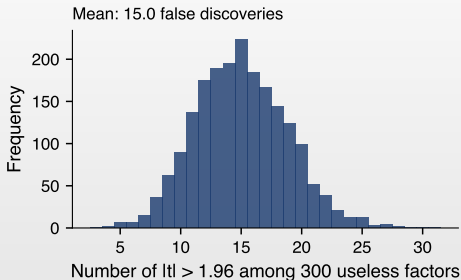
and reject $H_{(1)}, \dots, H_{(j)}$

- ▣ Controls FDR at α for independent tests, or under positive regression dependence on the true nulls (Benjamini & Yekutieli (2001))
 - Under arbitrary dependence: the Benjamini–Yekutieli variant divides the thresholds by $\sum_{j=1}^M 1/j$
- ▣ Less strict than **FWER** (Family-Wise Error Rate) control by Bonferroni or Holm (Holm (1979)), defined in Chapter 2: suited to searching many candidate factors

Worked Example: Ten Candidate Factors

- Sorted p -values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t -statistics from the standard Normal distribution (no true premium)

Reading the Simulation

- ▣ On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- ▣ On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications
- ▣ The median largest $|t|$ is 3.06: the best “factor” always looks strong
- ▣ Bonferroni threshold for 300 tests: $|t| > 3.76$
- ▣ Hurdles rise with the number of tests actually tried, including unpublished ones

Taming the Zoo and the Replication Debate

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add information given the existing zoo?
 - ▶ **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator): penalised regressions choose the controls among hundreds of factors
 - ▶ Most new factors are redundant; only a few add pricing information
- [Jensen, Kelly & Pedersen \(2023\)](#): a Bayesian view of 153 factors in 93 countries
 - ▶ Most factors replicate; they cluster into 13 themes; out-of-sample and international evidence is consistent
- The two views agree: the zoo is highly redundant, and a few themes carry the information

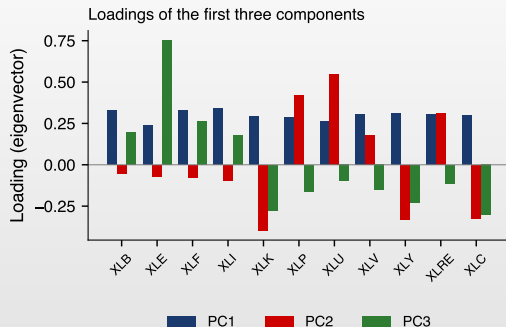
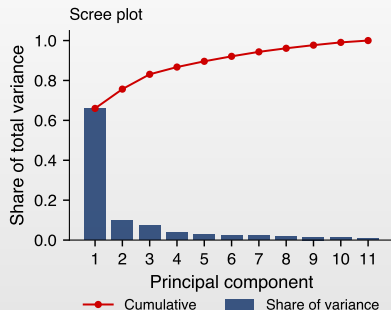
Recap: The Factor Zoo

- ▣ Hundreds of published factors imply many false discoveries at $|t| > 1.96$
- ▣ Bonferroni and Holm control the chance of any false discovery; BH controls their share
- ▣ With 300 useless factors, the best one has $|t| \approx 3$: raise the hurdle
- ▣ Redundancy is the main problem: few themes carry most of the information

Prerequisites Recap: Principal Component Analysis

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis) of the correlation matrix of standardised returns: $C = \Gamma \Lambda \Gamma^\top$, components $y_t = \Gamma^\top z_t$
- ▣ Eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$; eigenvectors are the **loadings**; component k explains λ_k/N of the variance
- ▣ Two assets with correlation ρ : $\lambda = 1 \pm \rho$, eigenvectors $(1, 1)/\sqrt{2}$ (the “market”) and $(1, -1)/\sqrt{2}$ (the spread); in a crisis ρ rises and PC1 dominates
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components; more examples in Seminar A8

PCA of the US Sector ETFs



11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)

Reading the Principal Components

- ▣ PC1 explains 66.0% of the variance; all loadings are positive (0.24–0.34)
 - ▶ Correlation of PC1 with SPY: 0.954; the first latent factor is the market
- ▣ PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology –0.40, consumer discretionary –0.33
 - ▶ A **defensive-versus-growth** factor, with no name given in advance
- ▣ PC3 (7.4%) is dominated by energy; three components explain 83.1%
- ▣ Only 2 eigenvalues exceed 1 (Kaiser rule): a heuristic, not a consistent estimator of the number of factors (next slides)

How Many Factors? Consistent Estimators

- Approximate factor model $x_{it} = \lambda_i^\top F_t + e_{it}$ with $N, T \rightarrow \infty$; $V(k)$: mean squared residual after the first k principal components
- Bai & Ng (2002): $\hat{k} = \arg \min_k \left[\ln V(k) + k \frac{N+T}{NT} \ln \min(N, T) \right]$ (criterion IC_{p2}), consistent as $N, T \rightarrow \infty$
- Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg \max_k \mu_k / \mu_{k+1}$, no penalty to choose
- Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- Kaiser ($\lambda_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Worked Example: The Number of Factors in Our Data

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size $\times$ B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k : 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N , so with small N the answer is fragile

Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- ▣ PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#); multivariate background: [Härdle & Simar \(2019\)](#)

Risk-Premium PCA and Three-Pass Premia

- **RP-PCA** (Risk-Premium PCA, [Lettau & Pelger \(2020\)](#)): principal components of $\frac{1}{T}X^T X + \gamma \bar{x}\bar{x}^T$
 - ▶ $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
 - ▶ In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- **Three-pass estimator** ([Giglio & Xiu \(2021\)](#)): the premium of an observed factor g_t when the model omits factors
 - ▶ (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- Both keep PCA's statistical backbone but target pricing, not only variance; IPCA (next) adds characteristics

Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of L observed characteristics $\mathbf{z}_{i,t}$ (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- ▣ $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?

What IPCA Teaches

- Title of the paper: “Characteristics are covariances”
- In their US stock data, a few IPCA factors price the cross section better than observable-factor models with as many factors
- The alpha term Γ_α becomes insignificant: characteristics predict returns because they measure risk exposure
- Links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins (Liu, Tsyvinski & Wu (2022)); dimension reduction on crypto and traditional return features: Pele et al. (2023), Jheng et al. (2026)

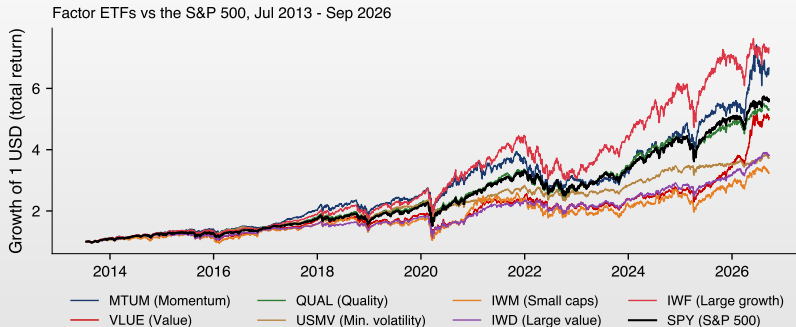
Recap: Latent Factors

- PCA finds factors as eigenvectors of the correlation matrix; choose their number with Bai–Ng or Ahn–Horenstein, not Kaiser
- In sector ETFs, PC1 is the market (66%), PC2 a defensive-versus-growth factor (10%)
- PCA explains variance, with fixed loadings; it is not an asset pricing model by itself
- IPCA lets loadings depend on characteristics and tests whether characteristics are risk

Factor ETFs

- ▣ **Smart beta:** rule-based portfolios that tilt towards factors, sold as ETFs
- ▣ MTUM (momentum), VLU (value), QUAL (quality: high profitability, low leverage)
- ▣ USMV (minimum volatility: the low-risk anomaly), IWM (small caps, Russell 2000)
- ▣ IWD / IWF: Russell 1000 value / growth (large caps)
- ▣ Long-only: they keep market exposure and tilt towards selected factors; they need not replicate the long leg of an academic factor or have a market beta close to 1

Factor ETFs versus the Market

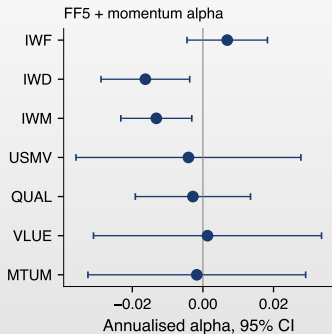


□ Growth of 1 USD with dividends re-invested, from 18 July 2013 (first common day)

Reading the Factor ETF Returns

- ▣ **CAGR:** large growth (IWF) 16.3% and momentum (MTUM) 15.5% beat SPY (14.0%)
- ▣ Quality 13.5%, value 13.1%, large value 10.7%, minimum volatility 10.5%, small caps 9.3%
- ▣ This period favoured large growth stocks: value and small caps lagged
- ▣ Thirteen years of one realisation: not evidence against the premia on its own

Alphas of Factor ETFs



Factor loadings

	Mkt-RF	SMB	HML	RMW	CMA	MOM
MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00

□ Daily regressions on FF5 + MOM with Newey–West errors, each ETF on its own trading calendar (2000–2026)

 MFM_ch3_etf_alphas

Reading the ETF Alphas

- ▣ The loadings match the labels: MTUM on MOM 0.36, VLUU on HML 0.34, IWM on SMB 0.85
- ▣ Growth (IWF) loads -0.28 on HML; USMV has market beta 0.74 and positive RMW, CMA
- ▣ Factor-model alphas are close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%
- ▣ Russell ETFs: IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.55$): costs and index construction
- ▣ R^2 between 0.81 and 0.97: the ETFs are mostly repackaged factor exposure

Factor Timing and Crowding

- **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
 - ▶ Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- **Crowding:** when many investors hold the same factor positions
 - ▶ Premia can shrink ([McLean & Pontiff \(2016\)](#)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers ([Daniel & Moskowitz \(2016\)](#))
- **Costs:** long-short factors ignore trading costs and short-selling fees; ETFs pay them

Recap: Factor Investing

- ▣ Factor ETFs deliver the promised exposures and keep most of the market exposure (betas 0.74–1.04)
- ▣ After controlling for the factors, their alphas are zero or slightly negative
- ▣ 2013–2026 rewarded growth and momentum, punished value and size
- ▣ Timing and crowding are open problems; evaluate them out of sample

Case Study: Research Question

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
 - ▶ No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
 - ▶ No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Question:** do published factors have significant CAPM alphas in the original data, out of sample and in other countries, once multiple testing is taken into account?
- **Why it matters**
 - ▶ If most factors are false discoveries, multifactor models and factor investing (Sections 6 and 9) rest on noise
 - ▶ The factor zoo (Section 7) is treated as a statistical problem; the paper treats it as correlated evidence on a few economic themes
- **Link to this chapter:** Jensen's alpha (Section 4) is the object tested; Benjamini–Yekutieli (Section 7) is the frequentist benchmark; Bayesian shrinkage extends the Vasicek beta (Seminar 3, A3) to alphas

Case Study: Data and Method

- **Data (Section II):** 153 factors in 93 countries; 138 from the list of [Hou, Xue & Zhang \(2020\)](#) plus 15 from the wider literature; US returns from 1926, most developed markets from 1986; all in US dollars, in excess of the US Treasury bill
 - ▶ Each country and month: terciles of the characteristic, breakpoints from non-micro stocks; "capped" value weights (market equity winsorised at the NYSE 80th percentile); long the tercile with the higher expected return in the original paper, one-month holding
 - ▶ At least 5 stocks per leg and 60 monthly observations; every factor scaled to a 10% annual idiosyncratic volatility
 - ▶ 13 themes: hierarchical clustering of US factors, distance 1 – correlation of CAPM residuals, Ward linkage (Section II.B)
- **Model (Section I):** $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$; alphas $\alpha_i = c_j + s_n + w_i$ (theme j , signal n , factor i ; eq. 21), prior mean 0
 - ▶ Prior variance $\Omega = MM^\top \tau_c^2 + ZZ^\top \tau_s^2 + I \tau_w^2$ (eq. 23); posterior mean $(\Omega^{-1} + T\Sigma^{-1})^{-1} T\Sigma^{-1} \hat{\alpha}$ (Proposition 4)
 - ▶ One factor: $E(\alpha | \hat{\alpha}) = \kappa \hat{\alpha}$, $\kappa = 1/(1 + \sigma^2/(\tau^2 T))$ (eq. 5): estimates from short, noisy samples are shrunk more
 - ▶ Empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\alpha} \sim N(0, \Omega + \hat{\Sigma}/T)$ (Section I.C.2)
- **Replication rate:** share of factors significant in the original paper whose alpha is significant here; Bayesian discovery if posterior mean / posterior s.d. ≥ 1.96 (eq. 26)

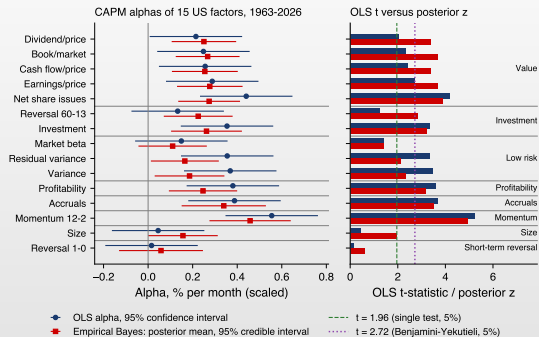
Case Study: Key Results

Replication rates (Figure 1)

Hou, Xue & Zhang (2020)	35.0%
US, raw return	55.6%
originally significant	61.3%
CAPM alpha	82.4%
Benjamini–Yekutieli	75.6%
Empirical Bayes	82.4%
World, empirical Bayes	82.4%

- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
 - ▶ $\Rightarrow$ testing alphas, not raw returns, drives most of the gap with Hou, Xue & Zhang (2020)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
 - ▶ $\Rightarrow$ shrinkage controls false discoveries without the power loss of a higher t hurdle
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size
 - ▶ $\Rightarrow$ not an artefact of micro-caps
- Mean alpha 0.49% a month in sample, 0.26% after the original sample: -47% ; slope of post-sample alphas on in-sample alphas 0.26 (Figure 8), against about 0.9 expected from shrinkage alone
 - ▶ $\Rightarrow$ decay is expected, but the strongest factors decay more: arbitrage or data mining
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

Case Study: Replication on Course Data



- 15 US long-short factors from value-weighted sorts of the [Kenneth French Data Library](#), July 1963 – July 2026 ($T = 757$ months); CAPM alphas scaled to 10% idiosyncratic volatility (Section II.A); themes of the paper ([authors' cluster labels](#)), prior mean 0

Case Study: Reading the Course Replication

- **Replication rates** (significant positive alpha): OLS 73.3% (11 of 15), Benjamini–Yekutieli 53.3% (8; critical $t = 2.72$), empirical Bayes 86.7% (13)
 - ▶ Benjamini–Yekutieli is the strictest, as in the US rates of the paper: 82.4%, 75.6%, 82.4% (Figure 1)
- **Shrinkage towards the theme**, not only towards zero: long-term reversal (investment theme) 0.13% $\rightarrow$ 0.23% a month ($t = 1.25$, posterior $z = 2.85$); net share issues 0.44% $\rightarrow$ 0.27%
 - ▶ Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- **Prior dispersion**: $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month, against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- **Themes**: the 15 signals fall into 8 of the paper's 13 themes (5 in value, 3 in low risk, 2 in investment); with a single common theme the empirical-Bayes rate is 100%
- **Failures**: market beta and short-term reversal ($z \leq 1.41$); size only just passes ($z = 1.96$), echoing the paper, where size is one of the two themes below 50%

Case Study: Replication, Limits and Extensions

- **Complete the course replication** (Seminar 3, C3): justify each sign from the original paper, compare the dendrogram with the paper's themes, add the Bayesian FDR and the post-sample slope (Figure 8)
- Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis
- **Limits stated by the authors**
 - ▶ CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
 - ▶ Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
 - ▶ Unobserved factors bias the prior: with the scenarios of [Harvey, Liu & Zhu \(2016\)](#) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Limits of the course version:** 15 US signals instead of 153 in 93 countries; library breakpoints instead of non-micro terciles with capped weights; one region, so $\tau_s = 0$
- **Open extension:** the paper's data end in 2020; did the posterior alphas and the replication rate hold in 2021–2026, and can a prior estimated on US factors sharpen short-sample BVB factors (Seminar 3, C1)?

An Open Question: Why Do Factor Premia Decay after Publication?

- ▣ McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication
 - ▶ the gap, 32%, is attributed to publication-informed trading; the out-of-sample drop bounds data mining
- ▣ Jensen, Kelly & Pedersen (2023): in a Bayesian model that shrinks in-sample premia, most of 153 factors replicate, in the US and in 93 countries
- ▣ Chen & Zimmermann (2022): open-source data and code reproducing the published cross-sectional predictors
- ▣ Open: how much of the decay is arbitrage, data mining or a change in risk, and does it differ across factors and markets?

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; check each headline number against the abstract

□ Hypothesis

- ▶ Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.

□ Data and code

- ▶ Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.
- ▶ Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).

▣ Critique

- ▶ Referee this: one factor, one publication date, $t = -1.45$. What is the power to detect a 58% decay, and why does pooling many anomalies change it?

▣ Report

- ▶ In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
- ▶ Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant enumerates specifications; the human fixes which ones count before running them

What the Researcher Must Check

- **Wrong event date:** sample end, working paper and journal issue are different dates
 - ▶ test: read them from the paper itself; Jegadeesh–Titman: sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay:** fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns:** Newey–West errors; rerun without 2009
- **Data vintages:** the factor library is revised retroactively
 - ▶ test: record the download date; reproduce the chapter's split-sample t -statistics
- **Misquoted literature:** a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power:** a single factor rarely rejects; report the confidence interval of the decay, not only its sign

Mini-Case: Was Momentum Arbitraged Away?

- A typical assistant proposal: “momentum disappeared once [Jegadeesh & Titman \(1993\)](#) published it”
- Design of [McLean & Pontiff \(2016\)](#): $MOM_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly, 1965–2026, Newey–West errors
 - ▶ post-sample: 1990-01 – 1993-03; post-publication: from 1993-04; $H_0 : b_2 = 0$ against $H_1 : b_2 < 0$
 - ▶ b_2 : total decline relative to the in-sample mean; $b_2 - b_1$: incremental effect of publication
- In-sample mean $a = 0.82\%$ a month ($t = 4.32$)
- Total decline $b_2 = -0.45\%$ ($t = -1.45$; 95% interval $[-1.05\%, 0.16\%]$): mean 0.37% a month, 54% lower
- Incremental $b_2 - b_1 = -0.79\%$ (s.e. 0.68%, $t = -1.18$); placebo, Mkt-RF: $b_2 = +0.40\%$ ($t = 1.12$)
- Interpretation: the point estimate is close to the 58% of [McLean & Pontiff \(2016\)](#), but one factor cannot reject H_0 at 5%
 - ▶ arbitrage is one interpretation, not an identified cause; pooling 97 predictors raises precision, while sample, cross-predictor dependence and heterogeneity also differ

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** do the Fama–French factors and momentum decay after publication, and does the decay differ across factors?
 - ▶ pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies (Mkt-RF: placebo, no publication date); errors robust to cross-factor correlation and autocorrelation
 - ▶ whether decay is larger where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs), not factor returns
- **Replicate first:** the chapter's split-sample t -statistics and the mini-case estimates
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , dates from each paper, factor set, corrections) committed before the results
 - ▶ code as a Quantlet; a report with Newey–West inference, placebo periods and power
- **Logs:** prompts in `AI_USE.md`; every caught error (a wrong publication date, a misquoted number) in `AI_ERRORS.md`

Conclusions

- ▣ The CAPM is elegant but fails on the 25 size $\times$ B/M portfolios: the SML is flat (slope -4.4%), $GRS = 4.20$
- ▣ Betas must be estimated with robust errors, change over time and should be shrunk
- ▣ Two-pass premia need errors robust to estimated betas and to misspecification; high R^2 on 25 portfolios is a low hurdle
- ▣ Hundreds of factors require multiple-testing hurdles ($|t| > 3$); models are compared by the Sharpe ratio of their factors
- ▣ Latent factors: PCA recovers the market and a defensive-versus-growth factor; IPCA links them to characteristics
- ▣ Factor ETFs are cheap factor exposure, not alpha

Links to Other Chapters

- ▣ **Chapter 2** (market efficiency): an alpha is either an anomaly (inefficiency) or a missing risk factor; the **joint hypothesis** problem
- ▣ **Chapter 4** (portfolio optimisation): factor covariance models and shrinkage for Markowitz
- ▣ **Chapters 5–6**: time-varying volatility and correlation, conditional betas
- ▣ **Chapter 8**: backtesting factor strategies without overfitting
- ▣ **Chapter 13**: machine learning for the cross section of returns

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- ▣ Factor data: [Kenneth French Data Library](#)

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